# Lecture Notes · Polynomials · Session 1 — K[x]: the other world with division

The construction of K[x] (with the honest formal subtlety), the degree rules proved, Euclidean division with existence and uniqueness in full, the ℤ[x] counterfactual, and the recycling of Euclid–Bézout with the explicit dictionary. Exercises and a Going further section at the end.

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## 1.1 The ring K[x]

**Definition 3.1.1.** Let K be a field (ℚ, ℝ, ℂ or ℤ/pℤ — our examples; the formal definition of field arrives in Linear Algebra). A **polynomial** over K is an expression

$$f = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, \qquad a_i \in K,$$

determined by its **list of coefficients** $(a_0, a_1, \ldots, a_n)$. Two polynomials are equal if they have the same coefficients. The set is $K[x]$, with addition (coefficient by coefficient) and multiplication (distribute and collect: the coefficient of $x^k$ in $fg$ is $\sum_{i+j=k} a_i b_j$).

> **What we will use from "field" (and where it gets fully defined).** Throughout the module $K$ is one of our familiar fields ($\mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{Z}/p\mathbb{Z}$), and of "being a field" we only exploit **two** properties — the two that carry the session's proofs: 1. **one can divide by any nonzero element** (every $a \neq 0$ has an inverse): used when dividing by the leading coefficient in Euclidean division (Theorem 3.1.4); 2. **there are no zero divisors** (a product of nonzero elements is nonzero): used in the degree rule (Proposition 3.1.3) and in "at most $n$ roots" (Session 2).
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> The complete **axiomatic definition** of field is given in Linear Algebra (M4 S1), when working over an arbitrary $K$ makes it indispensable; there it will land as a **summary** of what $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ and $\mathbb{Z}/p\mathbb{Z}$ already have in common. Here it suffices to name these two properties and use them rigorously.
>
> **Honest subtlety (form vs. function).** Here a polynomial is a **formal expression** (its list of coefficients), not the function it defines. Over ℝ the two can be identified without risk; over finite fields, **no**: in $\mathbb{Z}/2\mathbb{Z}$, the polynomials $x^2$ and $x$ are **different** (different coefficients) yet define **the same function** (they agree at 0 and 1). Equality of polynomials is equality of coefficients. *(The fine relation between the two — via "at most n roots" — in Session 2.)*

**Definition 3.1.2.** The **degree** $\deg f$ of $f \neq 0$ is the largest $n$ with $a_n \neq 0$; that $a_n$ is the **leading coefficient** (if it equals 1, $f$ is **monic**). The polynomial 0 has no degree (a convention).

**Proposition 3.1.3 (degree rules).** For $f, g \neq 0$: 1. $\deg(fg) = \deg f + \deg g$; in particular $fg \neq 0$ (**K[x] has no zero divisors**). 2. $\deg(f+g) \leq \max(\deg f, \deg g)$, with strict inequality only if the leading terms cancel.

*Proof.* (1) Let $a_m$ and $b_n$ be the leading coefficients of $f$ and $g$ ($m = \deg f$, $n = \deg g$), both $\neq 0$. The highest degree that can appear in $fg$ is $m+n$, and it is reached in **exactly one way**: multiplying the leading term of each factor. Hence the coefficient of $x^{m+n}$ in $fg$ is exactly the product $a_m b_n$ (no other summands feed it). **Here, and only here, the hypothesis enters:** to conclude $\deg(fg) = m+n$ we need that leading coefficient **not to vanish**; and in a field the product of two nonzero elements is nonzero — there are no **zero divisors**: if $ab = 0$ with $a \neq 0$, multiplying by $a^{-1}$ yields $b = 0$ (Arithmetic S3 proved it for $\mathbb{Z}/p\mathbb{Z}$; the argument is the same in any field). So $a_m b_n \neq 0$, the degree-$(m+n)$ term survives, $\deg(fg) = m+n$ and in particular $fg \neq 0$.

> **Where it would break without the hypothesis (the exact role of "no zero divisors").** The whole rule hangs on a single nail: that $a_m b_n \neq 0$. If the coefficients lived in a ring **with** zero divisors, that product of leading coefficients could vanish even with $a_m, b_n \neq 0$, and then the degree-$(m+n)$ term would cancel: **the degree of the product would drop** (it could even happen that $fg = 0$ with $f, g \neq 0$). Put differently, $\deg(fg) = \deg f + \deg g$ **is not a property of the polynomials, but of the field of coefficients** — and that is where "no zero divisors" pays off. Exercise 7 asks you to exhibit the failure in a concrete ring.

(2) Immediate, adding coefficient by coefficient: in degree $> \max(\deg f, \deg g)$ all coefficients are zero; strict inequality happens when the leading terms cancel (same degree and opposite leading coefficients). ■

> *Structural lens:* $K[x]$ is a **ring** — the same operations ($+$, $-$, $\times$) but **without requiring** every nonzero element to be invertible (like $\mathbb{Z}$ and $\mathbb{Z}/n\mathbb{Z}$; the name, without the theory). *Mind the hierarchy:* what we really used above is not "ring" plain and simple, but "**no zero divisors**" — an intermediate rung (a *domain*) that a field **guarantees** and an arbitrary ring does not ($\mathbb{Z}/6\mathbb{Z}$ is a ring and does have them). That is: **field $\Longrightarrow$ no zero divisors $\Longrightarrow$** (but not conversely) ring. And the view from the summit: nothing prevents forming $K[x][y]$ — polynomials in $y$ with coefficients in $K[x]$ — and asking again about its "primes", and again… **towers of rings**. Not course material; just so you know the landscape goes on (→ Going further).

## 1.2 Euclidean division

**Theorem 3.1.4 (division with remainder).** For $f, g \in K[x]$ with $g \neq 0$ there exist **unique** $q, r \in K[x]$ with

$$f = g\,q + r, \qquad \text{where } r = 0 \text{ or } \deg r < \deg g.$$

*Proof.* **Existence (strong induction on** $\deg f$ **— and it is, literally, the school long-division algorithm).** If $f = 0$ or $\deg f < \deg g$: take $q = 0$, $r = f$. If $\deg f = n \geq m = \deg g$, with leading coefficients $a$ and $b$: the polynomial

$$f_1 = f - \frac{a}{b}\,x^{n-m}\,g$$

cancels the leading term of $f$ (it was built to do so), so $f_1 = 0$ or $\deg f_1 < n$. **Here is where K being a field is used:** the step divides by the leading coefficient $b$ — one needs $b^{-1}$. By the induction hypothesis, $f_1 = gq_1 + r$ with $r$ admissible, and then $f = g(\frac{a}{b}x^{n-m} + q_1) + r$.

**Uniqueness ("subtract and bound degrees", as in $\mathbb{Z}$).** If $gq + r = gq' + r'$ with both remainders admissible: $g(q - q') = r' - r$. If $q \neq q'$, the left-hand side has degree $\geq \deg g$ (Proposition 3.1.3.1); the right-hand side is either 0 or has degree $< \deg g$. Impossible. Hence $q = q'$ and $r = r'$. ■

> **Example (the full long division, over $\mathbb{Q}$).** Let us divide $f = x^3 + x^2 + x + 1$ by $g = 2x + 1$ — the same divisor that will break $\mathbb{Z}[x]$ in 1.5, harmless here because in $\mathbb{Q}$ the 2 is invertible. At each step, divide the leading term of what remains by the leading term of $g$ and subtract that multiple:
>
> $$\begin{aligned} x^3 \div 2x = \tfrac{1}{2}x^2: &\qquad f - \tfrac{1}{2}x^2\,g = \tfrac{1}{2}x^2 + x + 1, \\ \tfrac{1}{2}x^2 \div 2x = \tfrac{1}{4}x: &\qquad (\tfrac{1}{2}x^2 + x + 1) - \tfrac{1}{4}x\,g = \tfrac{3}{4}x + 1, \\ \tfrac{3}{4}x \div 2x = \tfrac{3}{8}: &\qquad (\tfrac{3}{4}x + 1) - \tfrac{3}{8}\,g = \tfrac{5}{8} \quad (\deg < \deg g:\ \text{stop}). \end{aligned}$$
>
> The result is $q = \frac{1}{2}x^2 + \frac{1}{4}x + \frac{3}{8}$, $r = \frac{5}{8}$, and indeed $(2x+1)(\frac{1}{2}x^2 + \frac{1}{4}x + \frac{3}{8}) + \frac{5}{8} = x^3 + x^2 + x + 1$. Each step used $\frac{1}{2} = 2^{-1}$: **the inversion of the leading coefficient, three times**. The same computation in $\mathbb{Z}[x]$ gets stuck at the first step (there is no $\frac{1}{2}$) — that is exactly 1.5.

**Counterfactual 3.1.5 (the skeleton needs a body — a field).** In $\mathbb{Z}[x]$ (integer coefficients, not a field) division fails: dividing $x^2$ by $2x + 1$ requires, at the first step of the long division, the coefficient $\frac{1}{2} \notin \mathbb{Z}$ — and there are no $q, r \in \mathbb{Z}[x]$ with $x^2 = (2x+1)q + r$, $\deg r < 1$ (looking at leading coefficients: the leading coefficient of $q$ would have to be $1/2$). The exact step that breaks is the **inversion of the leading coefficient**, flagged in the proof of 1.4.

## 1.3 GCD, Euclid and Bézout: the dictionary

With division with remainder in hand, **all of §1.3–1.4 of Arithmetic replicates with the same proofs**, under the dictionary:

| In $\mathbb{Z}$ | In K[x] |
|---|---|
| absolute value | **degree** |
| the division of Theorem 1.1.3 (M1) | that of Theorem 3.1.4 |
| the remainder decreases ($0 \leq r < b$) | the **degree** of the remainder decreases |
| gcd = the greatest common divisor | gcd = the common divisor of **greatest degree**, normalized **monic** |

**Definition 3.1.6 (gcd in K[x]).** For $f, g \in K[x]$ **not both zero**, $d \in K[x]$ is a **greatest common divisor** of $f$ and $g$ if $d \mid f$, $d \mid g$ and every common divisor of $f, g$ divides $d$. Such a $d$ is determined up to a nonzero constant (the common divisors of greatest degree agree up to a factor); **normalizing it to be monic** makes it **unique**, and we write $\gcd(f, g)$. *(The convention "not both zero" avoids* $\gcd(0, 0)$*, which has no maximal degree.)*

**Theorem 3.1.7.** For $f, g \in K[x]$ not both zero the following hold, with the proofs from Arithmetic translated through the dictionary: 1. *(Euclid's lemma for the algorithm)* if $f = gq + r$, the common divisors of $(f, g)$ are those of $(g, r)$; the Euclidean algorithm terminates (degrees decrease — well-ordering) and produces the gcd. 2. *(Bézout)* $\gcd(f, g) = fu + gv$ for certain $u, v \in K[x]$, computable by working backwards (extended algorithm).

*Proof.* Those of Arithmetic (Lemma 1.1.6, Algorithm 1.1.7, Theorem 1.1.9 of those notes) **word for word**, substituting according to the dictionary; the only novelty is the monic normalization of the gcd (dividing by the leading coefficient — the field again) so that it is unique. Rewriting one of the three in full detail is exercise 6 — doing it once in your life is mandatory; twice, redundant. ∎

> **The module's message, already proved:** the proofs of Arithmetic never used anything about $\mathbb{Z}$ except division with remainder. This session adds no theorems: it adds the **awareness** that all that was a skeleton.

> **Example 3.1.8.** $\gcd(x^3 - 1,\ x^2 - 1)$ by Euclid: $x^3 - 1 = (x^2 - 1) \cdot x + (x - 1)$; $x^2 - 1 = (x - 1)(x + 1) + 0$. The gcd is $x - 1$ (already monic). Working backwards: $x - 1 = (x^3 - 1) \cdot 1 + (x^2 - 1) \cdot (-x)$ — explicit Bézout. *(Consistency check:* $x - 1$ *divides both, evident by factoring.)*

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> **Software (RA7) — division and factorization, done by the machine.** The first part is exactly Theorem 3.1.4: `div` returns the quotient and the remainder. The other two lines use results that arrive in sessions 2 and 3 —factoring, and counting roots with

multiplicity— so either leave them for then, or run them now and come back later to understand them.

```
from sympy import symbols, Poly, div, factor, roots
x = symbols('x')
f, g = Poly(x**4 - 1, x), Poly(x**2 + 1, x)
q, r = div(f, g)
print("quotient:", q.as_expr(), "| remainder:", r.as_expr())
print("factor x^4-1 over Q:", factor(x**4 - 1))
print("roots with multiplicity:", roots((x-2)**2 * (x+1)))
```

## Exercises

1. ★ Divide by long division, indicating at which steps you invert the leading coefficient: (a) $x^4 + 3x^2 - 1$ by $x^2 + x$, in ℚ[x]; (b) $x^3 + 2x + 1$ by $2x + 1$, in $\mathbb{Z}/\mathbf{5}\mathbb{Z}[\mathbf{x}]$ (the coefficients reduce mod 5; the inverse of 2 is 3!).
2. ★ Compute $\gcd(x^4 - 1,\ x^3 + x^2 + x + 1)$ by Euclid and work backwards for Bézout. Check the result by factoring both by eye.
3. ★ **(Counterfactual.)** Prove that there are no $q, r \in \mathbb{Z}[x]$ with $x^2 = (2x + 1)q + r$ and $\deg r < 1$. *(Look at the leading coefficient $q$ would need.)*
4. Complete the $\mathbb{Z} \leftrightarrow$ K[x] dictionary table with one more row: what plays the role of the numbers $\pm 1$ (the invertible elements of ℤ)? *(Reasoned answer: the nonzero constants — why exactly those?)*
5. Give two polynomials with $\deg(f + g) < \max(\deg f, \deg g)$ and state the exact condition for strict inequality.
6. Rewrite **in full detail** the proof of Bézout (Theorem 1.1.9 of Arithmetic) in its K[x] version: every occurrence of "smallest positive" must become "of smallest degree, monic". Point out the only spot where the translation requires thought.
7. Is it true that in ℤ/6ℤ[x] (coefficients in a **ring with zero divisors**) $\deg(fg) = \deg f + \deg g$ still holds? Give a counterexample. *(Hint: leading coefficients 2 and 3.)*

## Going further and references

**Going further (Euclidean domains, now with two examples).** ℤ and K[x] are the two canonical examples of a **Euclidean domain** (division with remainder relative to a "measure": absolute value / degree), and everything proved — Euclid, Bézout, and the unique factorization of Session 3 — holds in any of them. A third classic example: the Gaussian integers ℤ[i] (with the modulus from Complex Numbers as the measure!). Aluffi, *Algebra: Chapter 0*, ch. V; Childs.

**Going further (towers).** K[x,y] = K[x][y] has its irreducibles, but is **not** a Euclidean domain (there is no reasonable division with remainder in two variables) — and yet it retains unique factorization, by a theorem of Gauss. The landscape goes on, and it branches. Aluffi, ch. V.

**References.** Childs (division and Euclid in K[x], with the explicit parallel); Dorronsoro–Hernández, ch. 4; Hernández, *Álgebra y geometría*.

# Lecture Notes · Polynomials · Session 2 — Roots, the factor theorem and its geometry

The remainder and factor theorems with their proofs, the "at most n roots" with the fine detail (and the mod 8 counterfactual worked out), multiplicity well defined, the rational root theorem proved, and the geometric reading with its honest caveats. Exercises and a Going further section at the end.

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## 2.1 The remainder and factor theorems

**Definition 3.2.1.** $a \in K$ is a **root** of $f \in K[x]$ if $f(a) = 0$ (to evaluate: substitute $a$ for $x$ and operate in $K$).

**Theorem 3.2.2 (remainder theorem).** The remainder of dividing $f$ by $(x-a)$ is $f(a)$.

*Proof.* By Theorem 3.1.4, $f = (x-a)\,q + r$ with $r = 0$ or $\deg r < 1$: in both cases $r$ is a **constant**. Evaluating at $a$: $f(a) = 0 \cdot q(a) + r = r$. ■

**Corollary 3.2.3 (factor theorem).** $a$ is a root of $f \iff (x-a) \mid f$.

*Proof.* $(x-a) \mid f \iff$ the remainder is $0 \iff f(a) = 0$ (Theorem 3.2.2). ■

**Definition 3.2.4 (multiplicity).** Let $f \neq 0$ and let $a$ be a root of $f$. The **multiplicity** of $a$ in $f$ is the largest $m$ with $(x-a)^m \mid f$ (it exists and is finite: degrees bound it — for $f = 0$ there would be no such maximum, since $(x-a)^m \mid 0$ for every $m$). **Simple** root ($m = 1$), **double** ($m = 2$), **triple** ($m = 3$); and **multiple** when $m \geq 2$.

## 2.2 At most n roots (and where the hypothesis is used)

**Theorem 3.2.5.** A polynomial $f \neq 0$ of degree $n$ over a field $K$ has at most $n$ roots in $K$, **even counted with multiplicity**.

*Proof (**strong** induction on $n$ — the result is assumed for **every** degree $< n$ —, with multiplicities).* The induction has to be strong because the quotient appearing below has degree $n-m$ with $m \geq 1$ arbitrary, which may be much smaller than $n-1$. If $n = 0$, $f$ is a nonzero constant: zero roots. Let $n \geq 1$: if $f$

has no roots, the total is $0 \le n$, done. If $a$ is a root, let $m$ be its **multiplicity** (Definition 3.2.4); by maximality of $m$,

$$f = (x-a)^m\, q, \qquad q(a) \neq 0, \qquad \deg q = n - m$$

(Corollary 3.2.3 iterated and Proposition 3.1.3.1). Since $m \geq 1$, $\deg q = n - m < n$. Now let us see what happens with the **other** roots. Let $b \neq a$ be a root of $f$: from $0 = f(b) = (b-a)^m\, q(b)$, and since $(b-a)^m \neq 0$ and $K$ **has no zero divisors**, we get $q(b) = 0$ — that is, **every root of $f$ other than $a$ is a root of** $q$. Moreover, its multiplicity is preserved: since $(x-a)^m$ does not vanish at $b$, the factor $(x-b)$ only enters through $q$, so the multiplicity of $b$ in $f$ coincides with that of $b$ in $q$. By the induction hypothesis applied to $q$ (degree $n-m$), the sum of the multiplicities of the roots of $q$ is at most $n-m$. Adding the contribution of $a$:

$$\underbrace{m}_{\text{root } a} + \underbrace{(n-m)}_{\text{roots of } q} = n. \qquad \blacksquare$$

**Counterfactual 3.2.6 (the "field" hypothesis, portrayed).** In $\mathbb{Z}/8\mathbb{Z}$, the equation $x^2 = 1$ has **four** solutions: $1, 3, 5, 7$ (direct check: $3^2 = 9 \equiv 1$, $5^2 = 25 \equiv 1$, $7^2 = 49 \equiv 1$). A polynomial of degree 2 with 4 roots! What went wrong? The step flagged in the proof: from $(b-a)q(b) = 0$ we deduced $q(b) = 0$ using that there are no zero divisors — but in $\mathbb{Z}/8\mathbb{Z}$ there are ($2 \cdot 4 = 0$!): the factorization $x^2 - 1 = (x-1)(x+1)$ admits "crossed" roots such as $x = 3$: $(3-1)(3+1) = 2 \cdot 4 = 0$ **without either factor vanishing**. The dichotomy of Arithmetic S3, biting.

> **Conceptual consequence (closing the subtlety of S1):** over an **infinite** field, if two polynomials define the same function, they are the same polynomial (their difference would have infinitely many roots, impossible unless it is 0). Over finite fields we already saw this is not so — and now we know exactly why the argument does not apply.

## 2.3 Rational roots

**Theorem 3.2.7.** Let $f = a_n x^n + \cdots + a_1 x + a_0$ have **integer** coefficients, and let $p/q$ be a rational root written in lowest terms ($\gcd(p,q) = 1$). Then

$$p \mid a_0 \qquad \text{and} \qquad q \mid a_n.$$

*Proof.* From $f(p/q) = 0$, multiplying by $q^n$:

$$a_n p^n + a_{n-1} p^{n-1} q + \cdots + a_1 p\, q^{n-1} + a_0 q^n = 0.$$

*For $p \mid a_0$:* all summands except the last are multiples of $p$; hence $p \mid a_0 q^n$. Since $\gcd(p,q) = 1$, also $\gcd(p, q^n) = 1$ (**Arithmetic, Corollary 1.2.5** — Euclid's lemma — **iterated**: a common prime of $p$ and $q^n$ would divide some factor $q$), and now **Arithmetic, Lemma 1.2.4 (general version)** — if $\gcd(c, a) = 1$ and $c \mid ab$ then $c \mid b$ — gives $p \mid a_0$. *For $q \mid a_n$:* symmetric, and it is worth redoing in full because it is **the same pair of results again**: all summands except the **first** are multiples of $q$, hence $q \mid a_n p^n$; from $\gcd(q, p) = 1$ we get $\gcd(q, p^n) = 1$ (Corollary 1.2.5 iterated), and Lemma 1.2.4 gives $q \mid a_n$. ■

*(Citation warning: the two steps are **different** results from Arithmetic. Corollary 1.2.5 is "Euclid's lemma" proper — $p$ prime, $p \mid ab \Rightarrow p \mid a$ or $p \mid b$ — and is used only to propagate coprimality to powers; the one that delivers $p \mid a_0$ is Lemma 1.2.4, the **general version** with no primality hypothesis.)*

**Practical method (with its limits declared):** the candidates for rational roots are the quotients (divisor of $a_0$)/(divisor of $a_n$) — a **finite list**: test them and extract factors. It does not detect irrational or complex roots; it is a first filter, not a complete method.

**Corollary 3.2.8 (what the filter does say when not all roots come out).** Let $f \in \mathbb{Z}[x]$ and suppose **all** the candidates of Theorem 3.2.7 have been tested and the corresponding linear factors extracted, leaving $f = (\text{rational linear factors}) \cdot g$ with $\deg g \geq 2$. If no candidate annihilates $g$, then $g$ **has no rational root at all**: all of its roots are irrational or complex.

*Proof.* A rational root of $g$ would also be a root of $f$ (since $g \mid f$), hence it would be on the list of candidates of $f$ — and those have already been tested on $g$ without success. ■

> **Careful not to read too much into it.** "No rational roots" is **not** "irreducible over $\mathbb{Q}$": that implication only holds in degrees 2 and 3 (Proposition 3.3.3 of Session 3), and in degree 4 it breaks — $x^4 + 4$ has no rational roots and yet factors over $\mathbb{Q}$ as $(x^2 - 2x + 2)(x^2 + 2x + 2)$. What the filter delivers here is a statement about **roots**, not about factors.
>
> **Example (the filter in action).** For $f = 2x^3 - 3x^2 - 3x + 2$ the candidates are $\pm 1, \pm 2, \pm\frac{1}{2}$ (divisors of $a_0 = 2$ over divisors of $a_n = 2$; the $\frac{1}{2}$ is enabled by the leading coefficient 2). *Careful when counting:* crossing $\{\pm 1, \pm 2\}$ with $\{1, 2\}$ gives **eight** pairs, but $\pm 2/2$ is again $\pm 1$: **six** distinct candidates remain. Testing, $f(2) = f(-1) = f(\frac{1}{2}) = 0$, and $f = (x - 2)(x + 1)(2x - 1)$ — the complete factorization over $\mathbb{Q}$. The candidate $\frac{1}{2}$ illustrates the "$q \mid a_n$" half of the theorem: with leading coefficient 1 there would be no fractional roots.

## 2.4 The geometry of roots

**Over $\mathbb{R}$:** the real roots of $f$ are the points where the graph **touches the** $X$ **axis** — crossing it or not, as we see right away. Multiplicity can be *seen*: near a root $a$ of multiplicity $m$, $f(x) \approx c\,(x - a)^m$ (the other factors nearly constant), and $(x - a)^m$ **crosses** the axis if $m$ is odd and **bounces** (tangency) if $m$ is even. *(Honesty: "≈" and "near" are calculus language; the rigorous justification belongs to that course — here, the heuristic with the visual backing of GeoGebra.)*

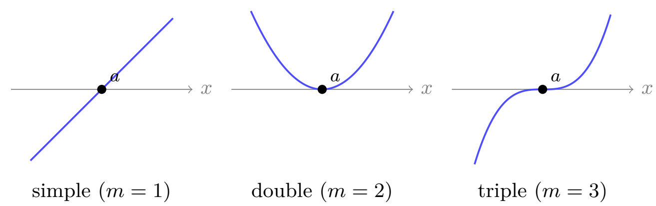



Figure 1: Three behaviours at the crossing $x = a$ according to the multiplicity $m$: simple ($m = 1$) crosses with nonzero slope; double ($m = 2$) touches and bounces without changing sign; triple ($m = 3$) crosses while flattening. Rule: odd $m$ crosses, even $m$ bounces.

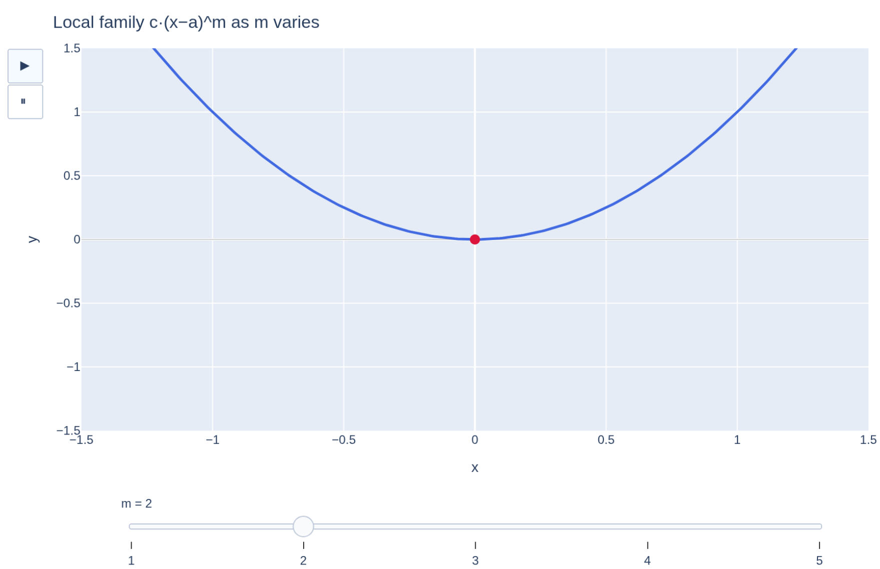
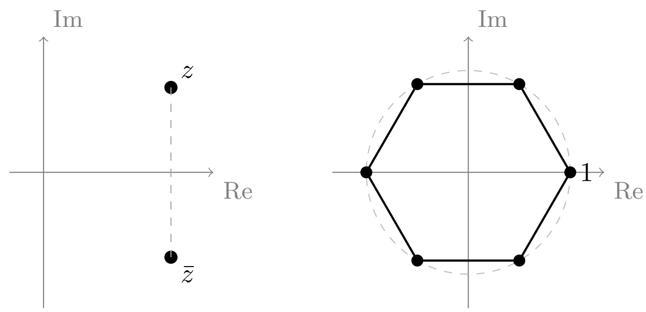



Figure 2: Local family $c\,(x-a)^m$ as $m = 1, 2, 3, \ldots$ varies: the transition crossing (odd $m$) ↔ bounce (even $m$) and the increasing flattening of the contact.

[see interactive version]

**Over $\mathbb{C}$:** the non-real roots of **real** polynomials come in conjugate pairs (Complex Numbers, Proposition 2.3.6 — cited, not reproved): mirror symmetry with respect to the real axis in the Argand plane. And the roots of $x^n - 1$ are exactly the **n-th roots of unity** (Complex Numbers, Theorem 2.3.1 with $w = 1$): the regular polygon, now reread as the *set of roots of a polynomial*. The bridge between the two modules, in a single figure.




(a) conjugate roots (b) roots of $x^6 - 1$

Figure 3: (a) A pair of conjugate roots $z, \bar{z}$ of a real polynomial, symmetric with respect to the real axis. (b) The roots of $x^n - 1$: the $n$ vertices of the regular polygon inscribed in the unit circle ($n = 6$, hexagon), with one vertex at 1.

> **The hypothesis "f real" is essential (and if the coefficients are complex?).** The conjugate symmetry **is not a property of the roots, but of the coefficients**. The proof relies on the fact that, when the $a_k \in \mathbb{R}$,
>
> $$\overline{f(z)} = \overline{\sum a_k z^k} = \sum \overline{a_k}\,\bar{z}^k = \sum a_k \bar{z}^k = f(\bar{z}),$$
>
> where the central step uses $\overline{a_k} = a_k$. Thus $f(z) = 0 \Rightarrow f(\bar{z}) = \overline{f(z)} = 0$. If **some** coefficient is not real, that step fails: what one obtains is $\overline{f(z)} = \bar{f}(\bar{z})$, with $\bar{f}$ **another** polynomial (the one with the conjugated coefficients), and the implication collapses. Minimal example: $f = x - i$ has the single root $i$, **not** $-i$; or $f = x^2 + i$, whose two roots are the negatives of each other, **not** conjugates. Moral: over general $\mathbb{C}[x]$ the roots need not come in mirror pairs — the symmetry is a gift exclusive to real coefficients. (It is precisely what, in Session 3, allows grouping the conjugate pairs into **real quadratic factors** when factoring over $\mathbb{R}$.)

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## Exercises

1. ★ The polynomial $f = x^3 - 2x^2 - 5x + 6$ has the root $x = 1$. Divide, lower the degree and factor completely over $\mathbb{Q}$.
2. ★ **(The filter, in both of its outcomes.)** (a) Apply Theorem 3.2.7 to $2x^3 + x^2 - 7x - 6$: list **all** the candidates, find the rational roots and

factor — here it comes out completely. (b) Now $2x^3 + x^2 + x - 1$: list the candidates, check that **only one** annihilates it, extract that linear factor and apply Corollary 3.2.8 to the remaining degree-2 quotient. What can you assert about its roots? And about its irreducibility over $\mathbb{Q}$?

3. ★ Sketch (and check in GeoGebra) the graph of $f = (x+1)(x-2)^2$: crossings, crossing vs. bounce (recall: 2 is a double root and $-1$ simple), and global behaviour.
4. ★ **(Counterfactual.)** Verify the four solutions of $x^2 = 1$ in $\mathbb{Z}/8\mathbb{Z}$ and, for $x = 3$, exhibit the pair of zero divisors that dodges the argument of Theorem 3.2.5. How many solutions does $x^2 = 1$ have in $\mathbb{Z}/p\mathbb{Z}$ with $p$ an odd prime, and why?
5. Build a monic polynomial of degree 5 with roots $-1$ (double) and 2 (triple), and decide how it crosses/bounces at each one.
6. Prove: if $f$ has degree $n$ and $n$ distinct roots $a_1, \dots, a_n$ are known, then $f = c(x - a_1)\cdots(x - a_n)$ for a constant $c$. *(Extract factors one by one, watching degrees.)*
7. Fill in a step that the proof of Theorem 3.2.5 uses without spelling it out: if $f = (x-a)^m q$ with $q(a) \neq 0$ and $b \neq a$, justify that the multiplicity of $b$ in $f$ is **exactly** that of $b$ in $q$. *(Look at how many times $(x-b)$ divides each factor.)*
8. True or false? "A polynomial of degree $n$ over a field has **exactly** $n$ roots in that field." Refute with two counterexamples of different natures (one over $\mathbb{R}$, another over $\mathbb{Q}$).
9. **(The hypothesis "f real".)** (a) Exhibit an $f \in \mathbb{C}[x]$ with a non-real root whose conjugate is **not** a root. (b) Locate the exact step of the conjugate-symmetry proof that uses $\overline{a_k} = a_k$ and explain what one obtains instead if some $a_k \notin \mathbb{R}$.

## Going further and references

**Going further (fast evaluation: Horner).** Evaluate $f(a)$ and divide by $(x-a)$ at the same time, with only $n$ multiplications: **Horner's scheme** (the school "Ruffini's rule" is its particular case). Useful and elegant; Childs, or any numerical methods text for its fine analysis.

**Going further (interpolation).** The practical converse of Theorem 3.2.5: through $n+1$ points with distinct abscissas passes **exactly one** polynomial of degree $\leq n$ (if two did, their difference would have $n+1$ roots). The explicit formula — **Lagrange** interpolation — reappears in numerical methods and in the FFT itself (a nod to the Going further of Complex Numbers). Hefferon or Childs.

**References.** Factor theorem and roots: Hernández; Childs. Rational roots: Childs. The graphical reading: any precalculus text, with GeoGebra as a laboratory.

# Lecture Notes · Polynomials · Session 3 — Unique factorization, irreducibles and the FTA in action

The module's closing act with all the proofs: unique factorization in K[x] (with the M1 dictionary executed), the complete factorizations of ℂ[x] and ℝ[x] (the FTA cashing in and the odd-degree promise settled), Vieta in general, and — as decided — the proofs of **Eisenstein and Gauss's lemma**, which in the videos are only stated. Exercises and a Going further section at the end.

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## 3.1 Irreducibles and unique factorization

**Definition 3.3.1.** A non-constant $f \in K[x]$ is **irreducible over K** if in every factorization $f = gh$ (in $K[x]$), $g$ or $h$ is constant. The nonzero constants are the **units** of $K[x]$ (the invertible elements — the role of $\pm 1$ in $\mathbb{Z}$; exercise 4 of S1). *The equality has two halves, and only one is substantial:* that every constant $c \neq 0$ **is** a unit is immediate, $c \cdot c^{-1} = 1$ — and there it pays off once more that $K$ is a **field**, which is what produces the $c^{-1}$; that there are **no others** follows from the degree rule ($fg = 1$ forces $\deg f = \deg g = 0$) and is exercise 4.

**Remark 3.3.2 (the dependence on the field).** $x^2 - 2$ is irreducible over $\mathbb{Q}$ (its roots $\pm\sqrt{2}$ are not rational and it has degree 2 — see Proposition 3.3.3) but factors over $\mathbb{R}$ as $(x-\sqrt{2})(x+\sqrt{2})$; $x^2+1$ is irreducible over $\mathbb{R}$ but not over $\mathbb{C}$, where $x^2+1 = (x-i)(x+i)$. **"Irreducible" without naming the field is an ill-posed question.**

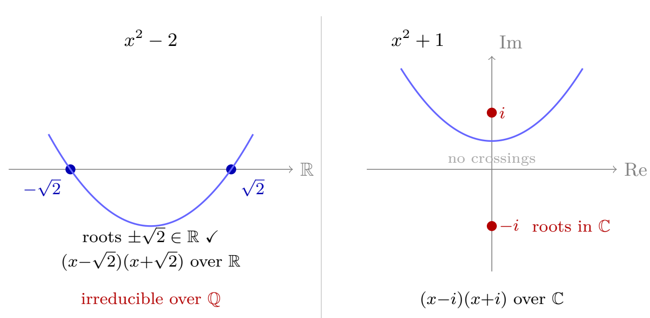


Figure 4: Irreducibility depends on the field: $x^2-2$ is irreducible over $\mathbb{Q}$ but factors over $\mathbb{R}$; $x^2+1$ is irreducible over $\mathbb{R}$ but factors over $\mathbb{C}$.

**Proposition 3.3.3 (criterion in degrees 2 and 3, and its limit).** A polynomial of degree 2 or 3 is irreducible over $K \iff$ it has no roots in $K$. **From degree 4 on the criterion is false:** $x^4 + 2x^2 + 1 = (x^2+1)^2$ has no real roots and is reducible over $\mathbb{R}$.

*Proof.* **($\Leftarrow$, by contraposition: if $f$ is reducible, it has a root.)** If $f = gh$ with both of smaller degree and $\deg f \in \{2,3\}$, some factor has degree 1 (the degrees add up to 2 or 3), say $ax+b$ with $a \neq 0$; then $-b/a$ is a root of $f$ (field: $a^{-1}$! — without an inverse of the leading coefficient the root cannot be solved for). **($\Rightarrow$, also by contraposition: if $f$ has a root, it is reducible.)** If $a \in K$ is a root, the factor theorem (Corollary 3.2.3) gives $f = (x-a)\,h$, and $\deg h = \deg f - 1 \in \{1,2\}$: **both factors are non-constant**, so $f$ is reducible. *(This is where one sees why degree 1 is left out of the statement: if $\deg f = 1$ the cofactor $h$ would be constant and the factorization trivial — a degree-1 polynomial is irreducible* ***and*** *has a root, so for it the $\iff$ is false.)* The degree-4 counterexample is verified by multiplying out. ■

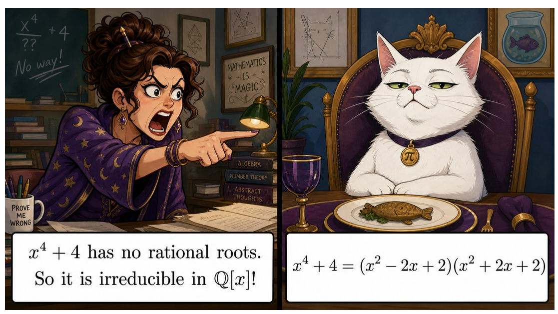


Figure 5: The classic mistake, so it is never forgotten: "no rational roots $\Longrightarrow$ irreducible" **only holds in degrees 2 and 3** — from degree 4 on, a polynomial can factor into quadratics without having any root at all.

**Theorem 3.3.4 (Euclid's lemma in K[x]).** If $p$ is irreducible and $p \mid fg$, then $p \mid f$ or $p \mid g$.

*Proof.* The one from Arithmetic (Lemma 1.2.4 + Corollary 1.2.5), dictionary in hand: if $p \nmid f$, then $\gcd(p,f) = 1$ (the divisors of $p$ are units or associates of $p$, by irreducibility — and $p \nmid f$ rules out the latter); Bézout in $K[x]$ (Theorem 3.1.7.2): $1 = pu + fv$; multiplying by $g$: $g = pug + (fg)v$, and $p$ divides both summands. ■

**Theorem 3.3.5 (unique factorization in K[x]).** Every non-constant polynomial of $K[x]$ is a product of irreducibles, uniquely up to order and constants (equivalently: fully unique if written $f = c \cdot p_1 \cdots p_r$ with $c \in K$ and the $p_i$ **monic** irreducibles).

*Proof.* **Existence:** strong induction on the degree, as in the fundamental theorem of arithmetic (Arithmetic): if $f$ is irreducible, done; if not, $f = gh$ with smaller degrees and induction. That produces $f$ as a product of irreducibles **plain and simple**; to reach the normalized form of the statement one **pulls each irreducible's leading coefficient out** — dividing by it makes the factor monic without changing its irreducibility (field again!) — and all those leading coefficients accumulate in the constant $c$ up front. It is that step, and no other, that turns "unique up to constants" into "fully unique". **Uniqueness:** as in Arithmetic (Theorem 1.2.6, the fundamental theorem), cancelling with Theorem 3.3.4: if $c\,p_1 \cdots p_r = c'\,q_1 \cdots q_s$ (monic irreducibles), $p_1$ divides the right-hand product, hence some $q_j$ (Theorem 3.3.4 iterated); both being monic irreducibles, $p_1 = q_j$; cancel and apply induction. Comparing leading coefficients gives $c = c'$. ■

## 3.2 The factorizations of $\mathbb{C}$[x] and $\mathbb{R}$[x]

**Theorem 3.3.6 (over $\mathbb{C}$, the FTA cashes in).** The irreducibles of $\mathbb{C}[x]$ are exactly those of degree 1. Every $f$ of degree $n \geq 1$ factors completely:

$$f = c\,(x - z_1)(x - z_2) \cdots (x - z_n),$$

and therefore has **exactly $n$ roots in $\mathbb{C}$ counted with multiplicity**.

*Proof (induction on $n$, explicit).*

**Base ($n = 1$).** $f = a_1 x + a_0$ with $a_1 \neq 0$ can be written $f = a_1(x - (-a_0/a_1))$: a single linear factor, with $c = a_1$.

**Step ($n \geq 2$, assuming the result for degree $n-1$).** Since $f$ is not constant, the **FTA** (Complex Numbers, Theorem 2.3.7 — **cited**: its proof uses analysis and exceeds the course, and so it was declared) guarantees a root $z_1 \in \mathbb{C}$. By the **factor theorem** (Corollary 3.2.3 of S2), $z_1$ is a root $\iff$ $(x - z_1) \mid f$, so

$$f = (x - z_1)\,q, \qquad \deg q = n - 1$$

(the degree of $q$ comes from $\deg f = \deg(x - z_1) + \deg q$, Proposition 3.1.3.1). By the induction hypothesis $q = c\,(x - z_2) \cdots (x - z_n)$, and substituting:

$$f = c\,(x - z_1)(x - z_2) \cdots (x - z_n).$$

**The count is exact ($n$, no more, no less).** The factorization exhibits $n$ linear factors, so $f$ has **at least** $n$ roots counted with multiplicity (grouping the equal $z_i$, the sum of multiplicities is $n$). And it cannot have more: if $a$ were a root, $0 = f(a) = c\,(a - z_1) \cdots (a - z_n)$, and since $\mathbb{C}$ has **no zero divisors**, some factor $a - z_i$ vanishes, that is, $a = z_i$ — every root is already on the list (this is, in passing, the "at most $n$" of Theorem 3.2.5 attaining equality). Hence **exactly** $n$, counted with multiplicity.

**The irreducibles are those of degree 1.** Any $f$ of degree $\geq 2$ has, by the above, a factor $(x - z_1)$ of smaller degree: it is reducible. And those of degree 1 are trivially irreducible (they do not split into factors of smaller degree, which would be constants). ■

**Theorem 3.3.7 (over $\mathbb{R}$).** The irreducibles of $\mathbb{R}[x]$ are: those of degree 1, and those of degree 2 **without real roots** (negative discriminant). Every real polynomial factors as a product of linear factors and irreducible quadratics.

*Proof.* Let $f$ be real of degree $n \geq 1$. By **Theorem 3.3.6**, over $\mathbb{C}$: $f = c\prod(x - z_i)$, with $c \in \mathbb{R}$ (it is the leading coefficient of $f$). The real roots contribute real linear factors. The non-real ones come **in conjugate pairs with the same multiplicity** — two **different** claims, with different sources:

- *that $\bar{z}$ is a root* if $z$ is: **Complex Numbers, Proposition 2.3.6** (cited, already proved there — and its proof uses that the coefficients are real, see the callout in S2 §2.4). That proposition **says nothing about multiplicities**;
- *that $z$ and $\bar{z}$ occur the same number of times*: a separate result, proved just below. It is not decoration: **without it the pairing fails**. If $z$ had multiplicity 3 and $\bar{z}$ multiplicity 1, grouping the pairs would leave a non-real factor $(x - z)^2$ over and the real factorization would not close.

Grouping each pair:

$$(x - z)(x - \bar{z}) = x^2 - (z + \bar{z})\,x + z\bar{z} = x^2 - 2\mathrm{Re}(z)\,x + |z|^2,$$

a **real** polynomial (Complex Numbers, Proposition 2.1.4(2) for $z + \bar{z} = 2\mathrm{Re}\,z$, and Proposition 2.1.5 for $z\bar{z} = |z|^2$), irreducible over $\mathbb{R}$ (degree 2 without real roots: its roots are $z, \bar{z} \notin \mathbb{R}$; Prop. 3.3.3).

*That the multiplicities of $z$ and $\bar{z}$ coincide.* Let $m$ be the multiplicity of $z$ in $f$, seen in $\mathbb{C}[x]$: $(x - z)^m \mid f$. Conjugating coefficient by coefficient and using that those of $f$ are real ($\bar{f} = f$), $(x - \bar{z})^m \mid f$ as well; since $z \neq \bar{z}$, the factors $(x - z)^m$ and $(x - \bar{z})^m$ share no root, so their product divides $f$ and $\bar{z}$ has multiplicity **at least** $m$. The symmetric argument ($z = \bar{\bar{z}}$) gives the other inequality, so both multiplicities equal $m$. Equivalently, and more in the style of this proof: since $z, \bar{z}$ are roots and $g = (x - z)(x - \bar{z})$ is **real**, the Euclidean division of $f$ by $g$ **in** $\mathbb{R}[x]$ (Theorem 3.1.4, with $f, g \in \mathbb{R}[x]$) gives a real quotient $f_1$ and a real remainder; but $g \mid f$ in $\mathbb{C}[x]$ (both roots of $g$ are roots of $f$) and the complex quotient is unique, so the remainder is 0 and $f = g\,f_1$ with $f_1 \in \mathbb{R}[x]$. Repeating on $f_1$ (of smaller degree) extracts one factor $g$ for each common unit of multiplicity of $z$ and $\bar{z}$, whence the coincidence.

*The real factorization, written out.* Running through all the pairs — each non-real root together with its conjugate, which by the above occur equally often, so that **none is left over** — yields, with $a_1, \dots, a_k \in \mathbb{R}$ the real roots (repeated according to their multiplicity) and $q_1, \dots, q_s$ the irreducible real quadratics of the $s$ conjugate pairs:

$$f = c\,(x - a_1) \cdots (x - a_k)\; q_1 \cdots q_s, \qquad k + 2s = n,$$

where each $q_j = (x - z_j)(x - \bar{z}_j) \in \mathbb{R}[x]$. Every factor is real, so **this is a factorization inside** $\mathbb{R}[x]$. And along the way: there are no real irreducibles other than those listed, since any real irreducible divides one of these factors (unique factorization, Theorem 3.3.5). ■

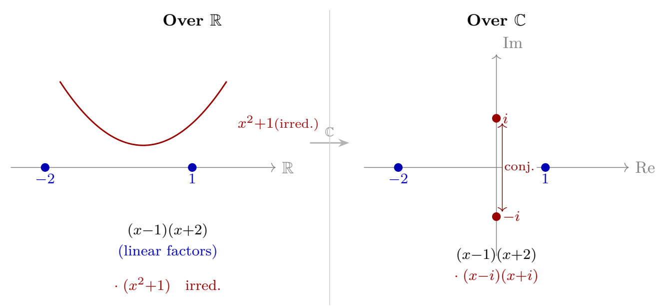


Figure 6: Over ℝ a polynomial factors into linear and irreducible quadratic factors; over ℂ those quadratics open up into pairs of conjugate roots.

**Corollary 3.3.8 (the promise from Complex Numbers, settled).** Every real polynomial of **odd degree** has at least one real root.

*Proof.* In the factorization of Theorem 3.3.7, $k + 2s = n$: the $s$ quadratic factors consume degree **two at a time**. If there were no linear factor ($k = 0$), the total degree $n = 2s$ would be even. Since $n$ is odd, $k \geq 1$: there is at least one factor $(x - a_1)$ with $a_1 \in \mathbb{R}$, that is, at least one real root. ■

**Proposition 3.3.9 (Vieta's formulas).** If $f = a_n x^n + \cdots + a_0$ has roots $z_1, \ldots, z_n$ (with multiplicity, in ℂ), then

$$z_1 + z_2 + \cdots + z_n = -\frac{a_{n-1}}{a_n}, \qquad z_1 z_2 \cdots z_n = (-1)^n \frac{a_0}{a_n}.$$

*Proof.* Expanding $f = a_n(x - z_1)\cdots(x - z_n)$ (Theorem 3.3.6): the coefficient of $x^{n-1}$ is $a_n \cdot (-(z_1 + \cdots + z_n))$ (choose the $x$ in all factors but one), and the constant term is $a_n(-z_1)\cdots(-z_n)$. Equate with $a_{n-1}$ and $a_0$. *(The intermediate coefficients give the sums of products of $k$ roots — the elementary symmetric functions; general statement in the Going further.)* ■

**What is reading the sum or the product good for? (applications, not just computation).** The value of Vieta is giving a symmetric quantity of the roots **without finding them**. (i) *Centering to solve:* in $x^3 + bx^2 + cx + d$ the substitution $x = y - b/3$ removes the quadratic term. The $b/3$ does not fall from the sky: by Vieta $\sum z_i = -b$, so the **mean** of the roots is $\frac{1}{3}\sum z_i = -b/3$, and the substitution $x = y - b/3$ sends each root $z_i$ to $z_i + b/3 = z_i - (\text{mean})$ — that is, it **subtracts the mean from each root**, centering them at their center of mass. It is the first step of Cardano's method (M2, Going further). (ii) *The product makes the rational filter **plausible**:* the constant term is, up to sign and leading coefficient, the product of the roots, $a_0 = (-1)^n a_n \prod z_i$; it is then natural that the numerators of the rational roots hide inside $a_0$. **Careful: this is not a proof of $p \mid a_0$** and does not replace the one in Theorem 3.2.7 — the product of the roots is not an integer in general, and the fine step is arithmetic (Arithmetic, Lemma 1.2.4). It is where the intuition lives, not where the proof comes from. (iii) *The incomputable, made accessible:* for degree $\geq 5$ there is no formula in radicals (Abel–Ruffini, Going further), but the sum, the product and **every** symmetric function of the roots can be read off instantly from the coefficients. *(Numerical verification in a concrete case: exercise 6; inverse use — reconstructing —: exercise 3.)*

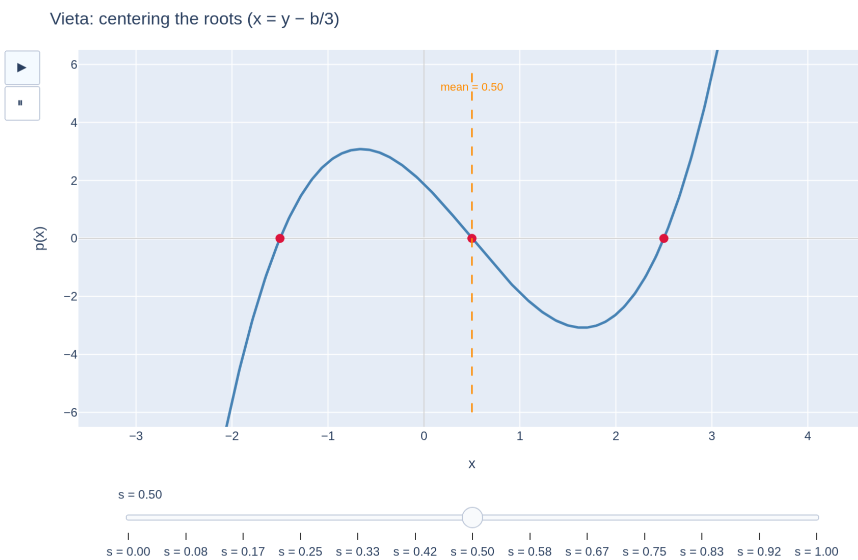


Figure 7: The substitution $x = y - b/3$ centers the roots at their mean (removes the quadratic term) — the first step of Cardano's method.

[see interactive version]

## 3.3 Eisenstein and Gauss's lemma (the promised proofs)

*(Design decision: in the videos they are stated and applied; here they are proved. Both proofs reuse $\mathbb{Z}/p\mathbb{Z}$ — Arithmetic paying off once more.)*

**Definition 3.3.10.** $f \in \mathbb{Z}[x]$ is **primitive** if the gcd of its coefficients is 1.

**Lemma 3.3.11 (Gauss).** The product of primitive polynomials is primitive.

*Proof.* Let $f, g$ be primitive and suppose $fg$ is not: some prime $p$ divides **all** the coefficients of $fg$. Reduce the coefficients modulo $p$ (write $\tilde{f}$ for the resulting polynomial in $\mathbb{Z}/p\mathbb{Z}[x]$; the reduction respects sums and products — it is operating with classes, Arithmetic S3). Then $\tilde{f} \cdot \tilde{g} = \widetilde{fg} = 0$ in $\mathbb{Z}/p\mathbb{Z}[x]$. But $\mathbb{Z}/p\mathbb{Z}$ is a **field**, so $\mathbb{Z}/p\mathbb{Z}[x]$ has no zero divisors (Proposition 3.1.3.1): $\tilde{f} = 0$ or $\tilde{g} = 0$ — that is, $p$ divides all the coefficients of $f$ or all those of $g$, against primitivity. ■

**Definition 3.3.11.1 (content).** For $f \in \mathbb{Z}[x]$, $f \neq 0$, its **content** $c(f)$ is the (positive) gcd of its coefficients, so that $f = c(f)\, f_0$ with $f_0 \in \mathbb{Z}[x]$ **primitive** (Definition 3.3.10). Thus "primitive" is exactly "$c(f) = 1$".

**Lemma 3.3.11.2 (multiplicativity of the content).** $c(fg) = c(f)\, c(g)$ for nonzero $f, g \in \mathbb{Z}[x]$.

*Proof.* Write $f = c(f)\, f_0$ and $g = c(g)\, g_0$ with $f_0, g_0$ primitive. Then $fg = c(f)c(g)\, f_0 g_0$, and $f_0 g_0$ is primitive by Lemma 3.3.11 (Gauss). Taking the content of both sides, $c(fg) = c(f)c(g)\, \underbrace{c(f_0 g_0)}_{=1} = c(f)c(g)$. ■

**Corollary 3.3.12 (the $\mathbb{Z} \leftrightarrow \mathbb{Q}$ step).** If $f \in \mathbb{Z}[x]$ factors in $\mathbb{Q}[x]$ into factors of degrees $r, s \geq 1$, then it already factors in $\mathbb{Z}[x]$ into factors of those same degrees.

*Proof.* Let $f = GH$ with $G, H \in \mathbb{Q}[x]$, $\deg G = r$, $\deg H = s$. Clearing denominators and extracting the content, each rational factor can be written $G = \frac{a}{b}\, G_0$ and $H = \frac{a'}{b'}\, H_0$ with $G_0, H_0 \in \mathbb{Z}[x]$ **primitive** (same degrees $r, s$) and rational scalars. Collecting the scalars into a fraction in lowest terms $\frac{u}{v}$ (with $\gcd(u, v) = 1$):

$$f = \frac{u}{v}\, G_0 H_0, \qquad \text{that is} \qquad v\, f = u\, G_0 H_0 \quad \text{in } \mathbb{Z}[x].$$

Taking contents on both sides (the content scales by the integer factor): $v\, c(f) = u\, c(G_0 H_0)$, and since $G_0 H_0$ is primitive (Lemma 3.3.11), $c(G_0 H_0) = 1$, so $v\, c(f) = u$. Hence $v \mid u$; but $\gcd(u, v) = 1$, so $v = 1$ and the scalar $\frac{u}{v} = u = c(f) \in \mathbb{Z}$. Consequently

$$f = \big(c(f)\, G_0\big)\, H_0,$$

a factorization **in** $\mathbb{Z}[x]$ with factors of degrees $r$ and $s$. ■

**Theorem 3.3.13 (Eisenstein's criterion).** Let $f = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$ and let $p$ be a prime with: $p \nmid a_n$; $p \mid a_i$ for all $i < n$; $p^2 \nmid a_0$. Then $f$ is **irreducible over** $\mathbb{Q}$.

*Proof.* Suppose $f = gh$ over $\mathbb{Q}$ with $\deg g = r \geq 1$, $\deg h = s \geq 1$; by Corollary 3.3.12 we may take $g, h \in \mathbb{Z}[x]$. Reduce modulo $p$: by hypothesis $\tilde{f} = \tilde{a}_n x^n$ (only the leading term survives), so $\tilde{g}\tilde{h} = \tilde{a}_n x^n$ in $\mathbb{Z}/p\mathbb{Z}[x]$. In this ring factorization is unique (Theorem 3.3.5 — $\mathbb{Z}/p\mathbb{Z}$ is a field!) and the only irreducibles dividing $x^n$ are associates of $x$: therefore $\tilde{g} = b\, x^r$ and $\tilde{h} = c\, x^s$ (nonzero constants $b, c$).

Since $r, s \geq 1$, the **constant terms of $g$ and $h$ are both divisible by $p$** — and then $a_0 = g(0)h(0)$ is divisible by $p^2$, against the hypothesis. ■

**Remark 3.3.13b (Eisenstein is sufficient, not necessary — and concludes nothing when it fails).** The criterion runs one way only: if a prime satisfying the three conditions **exists**, $f$ is irreducible. If **none exists**, nothing has been proved, and both outcomes genuinely occur:

- $x^4+4$ admits **no** Eisenstein prime (one would need $p \mid 0$ and $p \mid 4$ with $p^2 \nmid 4$, and no prime does that) and **is reducible**: $x^4+4 = (x^2-2x+2)(x^2+2x+2)$ (exercise 2(c));
- $x^2+1$ admits none either (one would need $p \mid a_0 = 1$, impossible) and yet **is irreducible over** $\mathbb{Q}$: degree 2 without rational roots (Prop. 3.3.3). Likewise $x^2+x+1$.

Moral: "there is no Eisenstein prime" is **no** evidence of reducibility, nor of irreducibility. When the criterion does not apply, one must decide by another route.

> **Example 3.3.14.** $x^4+10x+5$ is irreducible over $\mathbb{Q}$ ($p=5$). And $x^n-2$ is for **every** $n$ ($p=2$): over $\mathbb{Q}$ there are irreducibles of all degrees — what a difference from $\mathbb{R}$ (degree $\leq 2$) and $\mathbb{C}$ (degree 1)! The difficulty of irreducibility **lives in the field**.

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## Exercises

1. ★ Factor $x^4-4$ completely over $\mathbb{Q}$, over $\mathbb{R}$ and over $\mathbb{C}$ (three different answers — Remark 3.3.2, lived).
2. ★ Decide with Eisenstein (naming the prime): (a) $x^5+6x^3+12$; (b) $2x^7+9x^2+3$; (c) $x^4+4$ — careful: here Eisenstein **does not apply** with any p; decide its (ir)reducibility over $\mathbb{Q}$ by another route *(hint: $x^4+4 = (x^2+2)^2-(2x)^2$).*
3. ★ A monic real polynomial of degree 4 has roots $1+i$ and 2. If the fourth root is real and the constant term equals $-8$, reconstruct it (Vieta + conjugate pairs) and factor it in $\mathbb{R}$[x].
4. Prove that the units of $K[x]$ are exactly the nonzero constants. *(Degrees: $\deg(fg)=0$ forces…)*
5. Build your own counterexample to the "root criterion" in degree 4 over $\mathbb{Q}$ (no rational roots and reducible over $\mathbb{Q}$).
6. Verify Vieta for $2x^3-3x^2-11x+6$ (roots: $3, \frac{1}{2}, -2$) — sum and product against coefficients.
7. Write the complete proof of **existence** in Theorem 3.3.5 (strong induction on the degree), comparing it line by line with that of the fundamental theorem of arithmetic (Arithmetic).
8. **(The content in action.)** (a) Compute $c(f)$, $c(g)$ and $c(fg)$ for $f = 6x^2+4$, $g = 9x+3$, and verify Lemma 3.3.11.2. (b) For $f = 2x^2-2$, which factors in $\mathbb{Q}[x]$ as $(\frac{2}{3}x-\frac{2}{3})(3x+3)$, follow the proof of Corollary 3.3.12 step by step (identify $G_0, H_0, \frac{u}{v}$) until you obtain a factorization in $\mathbb{Z}[x]$.

9. Is $x^2 + 1$ irreducible over $\mathbb{Z}/5\mathbb{Z}$? And over $\mathbb{Z}/3\mathbb{Z}$? *(Degree 2: it suffices to look for roots — Prop. 3.3.3 — testing the 5 and 3 values.)*

## Going further and references

**Going further (symmetric functions and more Vieta).** The intermediate coefficients of $a_n \prod(x - z_i)$ are, up to sign, the **elementary symmetric functions** of the roots (sums of products taken $k$ at a time). The fundamental theorem of symmetric functions — every symmetric expression in the roots can be written in terms of the coefficients — is the historical antechamber of Galois theory. Aluffi, ch. VII; Childs.

**Going further (the skeleton has a name: Euclidean domains).** What $\mathbb{Z}$ and $K[x]$ share — division with remainder relative to a *measure* (absolute value / degree), and from there Euclid, Bézout and unique factorization — is the definition of a **Euclidean domain**. Note the noun: a **domain**, not just a ring — a commutative ring **without zero divisors**, the intermediate rung of the hierarchy *field* $\Longrightarrow$ *domain* $\Longrightarrow$ *ring* flagged in Session 1. Without that condition neither the degree rule nor "at most $n$ roots" would hold. A third classic example: the Gaussian integers $\mathbb{Z}[i]$, measuring with the modulus of complex numbers. The development (with the three examples and the branching towards $K[x, y]$, no longer Euclidean yet still factoring) is in the **Going further of Session 1**. Aluffi, ch. V; Childs.

**Going further (and the formulas?).** For degrees 2, 3 and 4 there are formulas in radicals (the school quadratic; Cardano — complex numbers were born there, recall the Going further of M2). For degree **5 and beyond, they do not exist** (Abel–Ruffini), and explaining *why* is exactly **Galois** theory — the natural destination for whoever enjoyed this module. Aluffi, ch. VII.

**References.** Factorization and irreducibles: Childs (the polynomial chapters — the closest reference); Dorronsoro–Hernández, ch. 4. Eisenstein and Gauss: Childs; Aluffi, ch. V. $\mathbb{C}$[x] and $\mathbb{R}$[x]: Hernández.